



UNIVERSITÀ DEGLI STUDI DI GENOVA

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DEPARTMENT OF INFORMATICS, BIOENGINEERING, ROBOTICS AND SYSTEMS ENGINEERING

ASSIGNMENT D3-V3

Warehouse Robotics: Heterogeneous Robots

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Chapter 1

Technical Report for Assignment D3-V3

Abstract

This report presents the modeling and resolution of an AI planning scenario involving a team of heterogeneous robots operating in a shared warehouse environment. The work addresses the so-called task-motion gap by combining two complementary planning formalisms: Classical Planning (PDDL), used to establish the logical dependency chains that govern task-level cooperation, and PDDL+, used to model continuous-time interactions and physical execution constraints. Taken together, the two models demonstrate how specialized transport and manipulation robots can be made to cooperate, by necessity, in order to complete logistics tasks across zones of the warehouse that neither robot type can traverse alone.

1.1 Introduction and Domain Overview

The domain simulates a warehouse populated by two functionally distinct classes of robots, each equipped with a complementary and mutually exclusive set of capabilities:

- **Manipulation robots (`manip-robot`):** comparatively slow units equipped with grippers, able to pick up, put down, and transfer packages. These robots are, however, strictly forbidden from entering designated transit areas of the warehouse.
- **Transport robots (`transport-robot`):** faster mobile platforms capable of navigating the entire warehouse, including restricted transit zones, but lacking any manipulation hardware.

Since no single robot in the fleet possesses both the spatial reach to cross the warehouse and the mechanical means to lift a package, the planning domain is constructed so as to force cooperation between the two robot types whenever a delivery task spans a transit zone. This design choice is central to the assignment: it transforms warehouse logistics from a single-agent path-planning problem into a genuinely multi-agent coordination problem.

1.2 Classical PDDL Model

1.2.1 Domain Modeling Choices

In the classical PDDL formulation, the physical transfer of a package between two robots is abstracted as an instantaneous change of logical state. The principal modeling decisions are summarized below.

- **Typing and predicates.** Robots are explicitly typed in order to prevent the planner from assigning capabilities to the wrong robot class. The predicates (`manip-holding`) and (`transport-carrying`) are kept distinct so that the physical state of a payload is always unambiguous with respect to which robot type is responsible for it.
- **Navigation constraints.** The warehouse topology is encoded through a (`transit-area ?1`) predicate. The `manip-move` action carries a hard negative precondition, (`not (transit-area ?12)`), which has the effect of confining manipulation robots to their designated zones and excluding them from transit areas altogether.
- **Mutual exclusion.** Coordination between concurrently active agents is enforced through an (`occupied ?1`) predicate, which acts as a lock on each spatial node. This guarantees that no two robots can ever occupy the same location simultaneously, thereby preventing physical collisions at the symbolic level.
- **Adjacent handoffs.** To avoid deadlocks in which two robots would otherwise need to occupy the same node at the same time to exchange a package, the `manip-load-onto-transport` and `unload` actions are defined to operate across pairs of adjacent nodes rather than a shared one. This allows a handoff to take place safely while each robot remains in its own location.

1.2.2 Problem Scenarios

Two problem instances were designed to validate the domain with increasing levels of complexity.

Problem 1: Single-Robot Delivery. This baseline scenario tests the domain in its simplest form: a single `manip-robot` operates within a small, fully connected area consisting of Shelf A and Shelf B linked by a corridor. As no transit-area restriction is present in this configuration, the manipulation robot is able to complete the delivery entirely on its own, picking up the package at Shelf A, moving through the corridor, and placing it down at Shelf B (Figure 1.1).

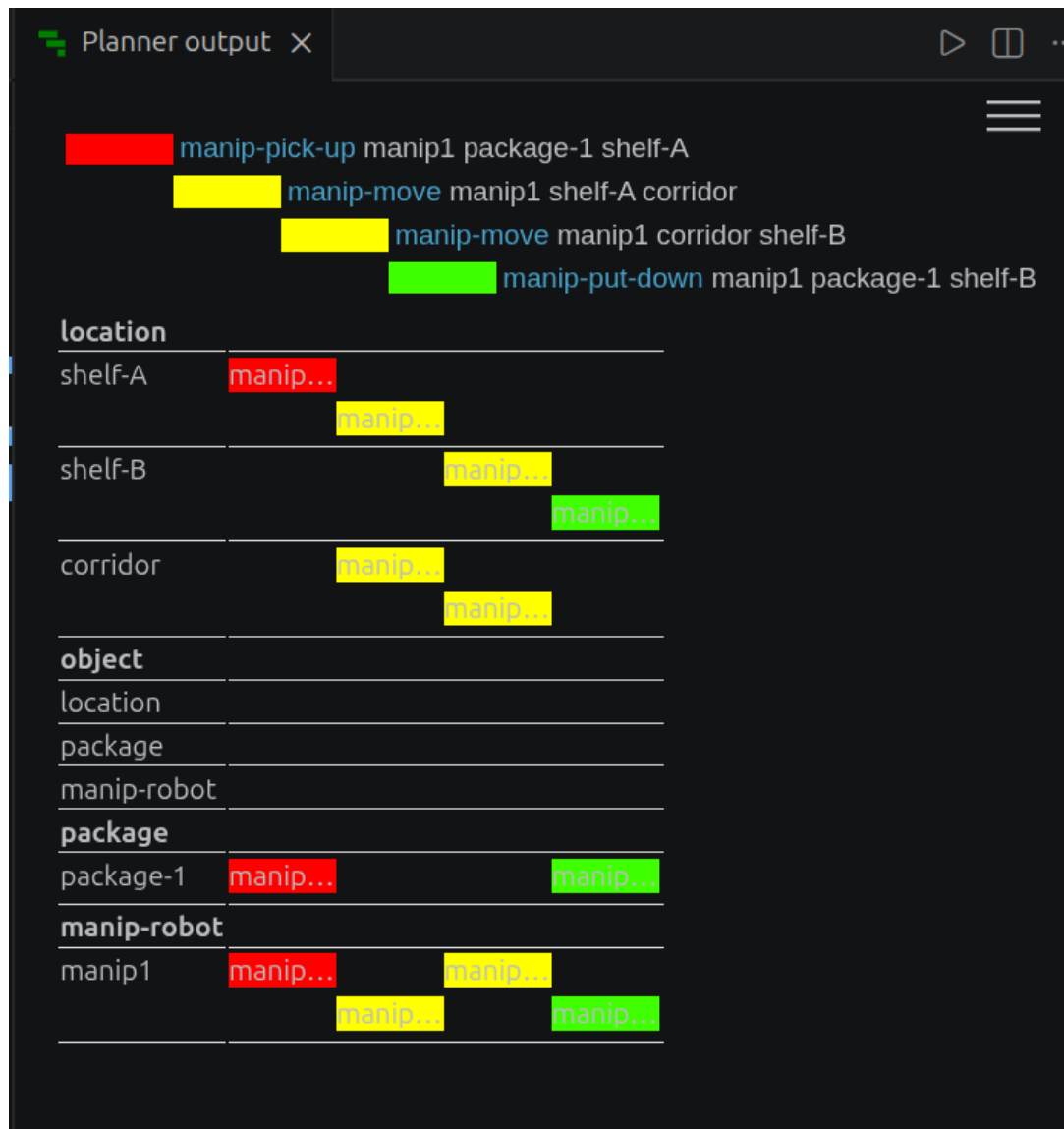


Figure 1.1: Setup and execution of the Single-Robot Delivery scenario.

Problem 2: Cooperative Delivery. This is the primary heterogeneous-cooperation scenario and the central test case of the assignment. The warehouse is partitioned into **zone-A** and **zone-C**, separated by a transit area that only transport robots may enter. **manip1** picks up the package in **zone-A** and loads it onto **transport1** at a designated handoff point. The transport robot then crosses the transit area and delivers the package to **manip2**, who completes the task by placing it down in the final zone (Figure 1.2). This scenario demonstrates that the planner correctly infers the need for inter-agent cooperation purely from the structure of the domain, without this cooperation being hard-coded into the problem instance.



Figure 1.2: Setup and execution of the Cooperative Delivery scenario across the transit area.

1.3 PDDL+ Continuous Time Model

1.3.1 Hybrid Dynamical Systems Integration

Real-world physical interactions are not instantaneous, and the classical model's treatment of package transfer as a single discrete event does not capture this. To bridge the gap between symbolic planning and continuous physical execution, the domain was extended with PDDL+ constructs capable of representing continuous time, numeric state variables, and dynamic safety thresholds.

- **Numeric fluents.** A continuous numeric variable, $(\text{transfer-progress } ?p)$, is

introduced to track the completion percentage of an in-progress transfer.

- **Continuous processes.** Once the `start-transfer` action locks both robots in place, a `transfer-process` runs continuously in the background, increasing the progress variable over time according to the differential equation `(* #t 10.0)`, i.e., progress accumulates at a constant rate of ten units per second.
- **Event-triggered failures.** An autonomous `transfer-timeout` event models hardware failure: if the transfer progress exceeds 100 without planner intervention, the package is dropped. Consequently, the planner must schedule the `finish-transfer` action precisely while progress lies between 80 and 99, balancing safety margin against process completion.

This construction obliges the planner to reason not only about which actions to execute, but also when to execute them relative to an autonomously evolving continuous process, which is the defining challenge of hybrid (discrete/continuous) planning.

1.4 Results and Execution

Both domains were tested using planners appropriate to their respective formalisms. The classical PDDL model, being purely discrete, can be solved with standard fast-forward planners. The PDDL+ model, in contrast, requires a planner capable of reasoning about numeric fluents and continuous processes.

Planner used: ENHSP (Expressive Numeric Heuristic Search Planner) was selected for the PDDL+ problem, as it correctly handles the non-linear numeric conditions and continuous processes involved, which standard temporal planners such as OPTIC are not designed to support.

1.4.1 Plan Validation

When solving `problem_q2_combined.pddl`, the planner correctly derives the required spatial cooperation strategy between the manipulation and transport robots. Crucially, during the handoff phase, the ENHSP output explicitly generates a waiting interval:

```
----- waiting - - - - [8.0]
```

This entry indicates that the planner deliberately allows 8.0 seconds to elapse while the `transfer-process` runs in the background, accumulating $8.0 \times 10.0 = 80$ units of progress. At precisely $t = 8.0$ s, the planner schedules `finish-transfer`, securing the package safely before the timeout event would otherwise trigger at $t = 10.0$ s. This confirms that the plan respects the hybrid continuous-time constraints with a comfortable, but not excessive, safety margin (Figure 1.3).

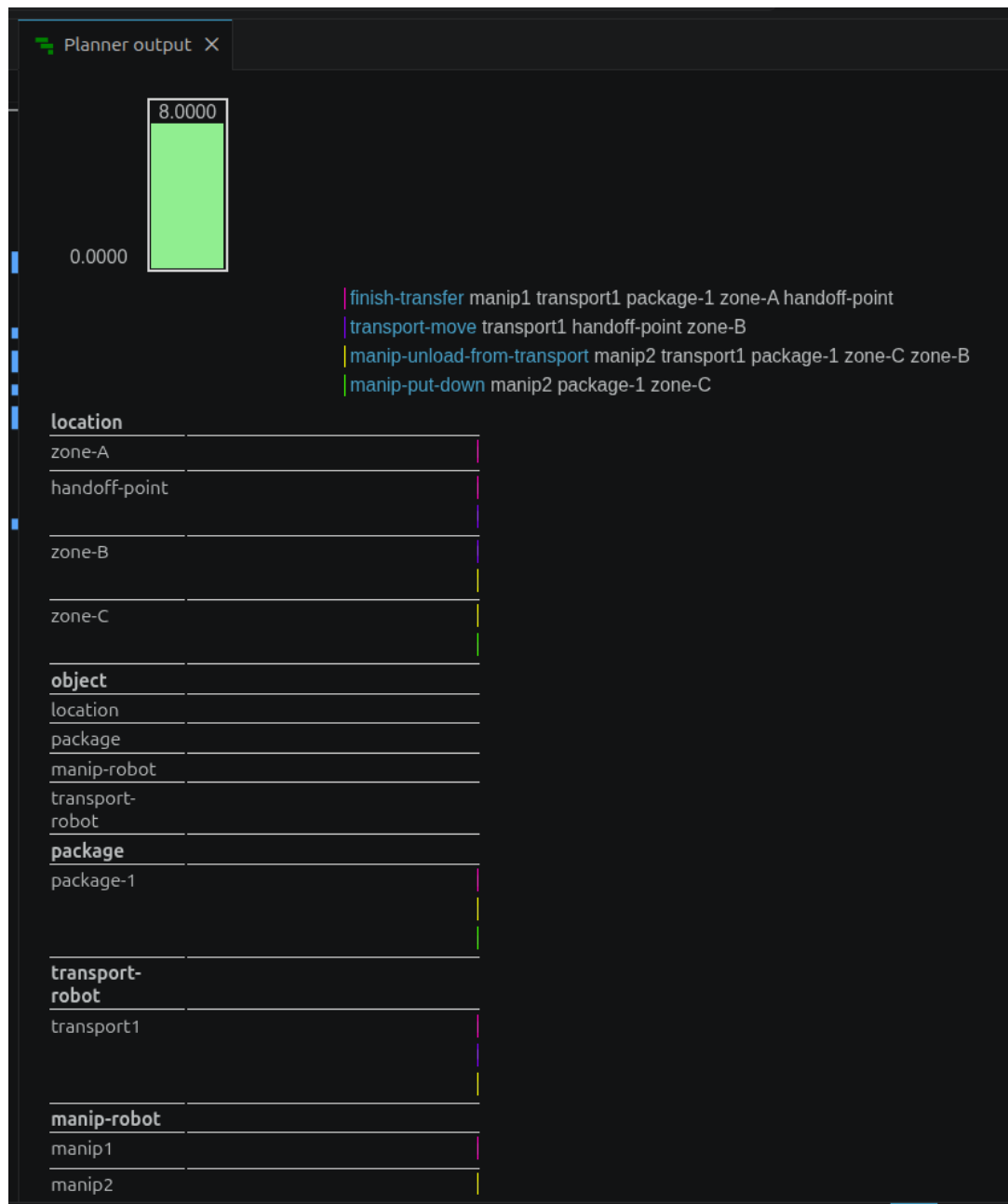


Figure 1.3: Execution trace from the ENHSP planner showing the scheduled wait of 8.0 seconds and the subsequent discrete actions that complete the handoff before the timeout threshold.

1.5 Discussion and Limitations

While both models successfully demonstrate heterogeneous multi-robot cooperation, several simplifying abstractions are worth highlighting, as they mark the boundary between what symbolic planning can represent and what a real deployment would require.

- 1. State-space scalability.** The introduction of PDDL+ processes substantially increases search complexity relative to the classical model. Extending the scenario to dozens of transport robots and packages operating concurrently would likely lead to a

combinatorial explosion in the state space, with a corresponding increase in planning time.

- 2. Deterministic time constraints.** The continuous transfer process assumes a perfectly linear, deterministic progression of time. In a real deployment, sensor noise, network latency, and physical slippage would introduce timing uncertainty that classical PDDL+ is not naturally equipped to represent.
- 3. Abstraction of collisions.** The *occupied* predicate resolves collisions at the topological (graph) level, but it abstracts away low-level 3D geometric constraints entirely — for instance, two large transport robots attempting to pass one another in a narrow corridor without physically colliding is a scenario the model cannot capture.

1.6 Conclusion

By assigning each robot class a strictly limited and complementary set of capabilities, and by introducing an impassable boundary in the form of the transit area, the domain successfully guarantees that heterogeneous multi-agent collaboration emerges as a structural necessity rather than an explicit instruction. The progression from the classical PDDL model to its PDDL+ extension illustrates a broader point about modern automated planning: moving from discrete logical sequencing to continuous physical dynamics substantially increases representational power, but does so at a non-trivial computational cost. Together, the two models offer a compact but informative case study in the trade-offs that characterize hybrid task-and-motion planning.